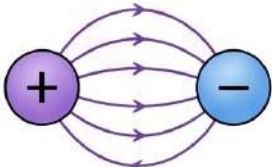


Electrostatics: Charge & Fields

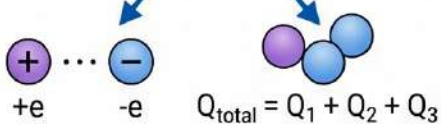
MHT-CET
Weightage: 7.4-8%

Electric Charge Basics

Charge is a property of matter due to which two bodies interact with each other electromagnetically.



Charge is quantized and additive.

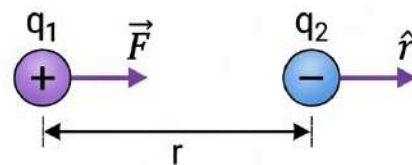


Coulomb's Law

Force between two charges:

$$\vec{F} = \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{q_1q_2}{r^2} \hat{r}$$

(where ϵ_r is dielectric constant and ϵ_0 is permittivity of free space)

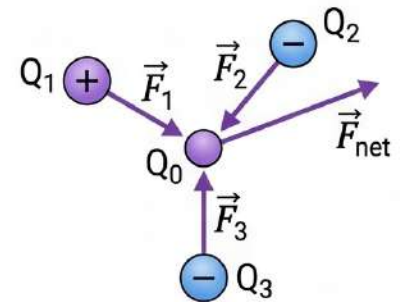


Coulomb's Law is applicable only for static point charges.

Principle of Superposition

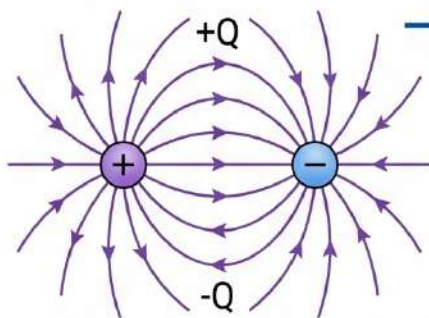
Force on a point charge due to many charges is given by

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots$$



The force due to one charge is not affected by the presence of other charges.

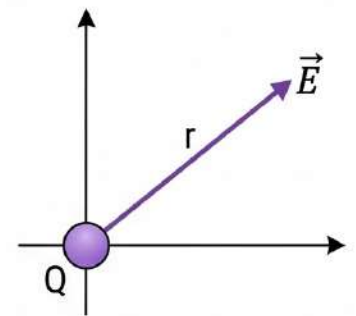
Electric Field Intensity ($\vec{E}$)



$$\vec{E} = \frac{\vec{F}}{q} \text{ with SI unit N/C or V/m.}$$

Electric Field due to Point Charge Q:

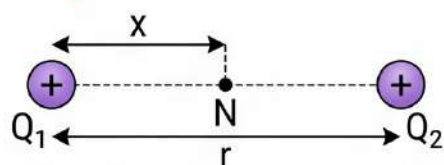
$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r}$$



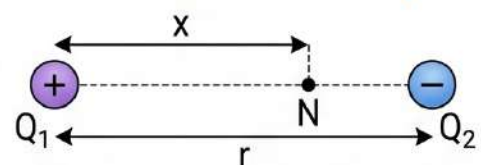
Null Point for Two Charges

$$\text{Distance } x \text{ of null point from } Q_1: x = \frac{\sqrt{Q_1}r}{\sqrt{Q_1} \pm \sqrt{Q_2}}$$

Use (+) for like charges and (-) for unlike charges.



Like Charges (e.g., $+Q_1, +Q_2$)

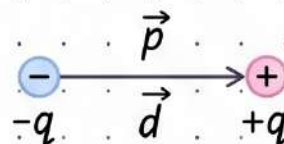


Unlike Charges (e.g., $+Q_1, -Q_2$)

Electric Dipole & Field Lines

Electric Dipole Basics:

Electric dipole moment $\vec{p} = q\vec{d}$, where $\vec{d}$ is distance from negative to positive charge.



Torque on dipole placed in a uniform electric field: $\vec{\tau} = \vec{p} \times \vec{E}$

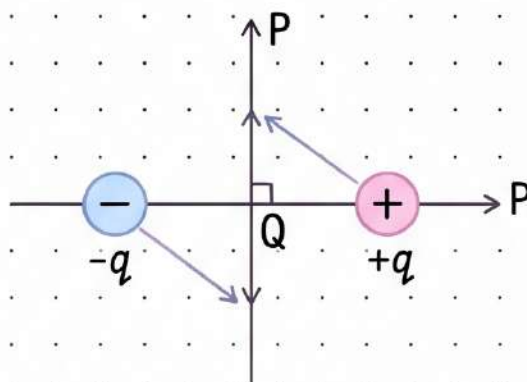
Field & Potential due to Dipole:

Axial Point:

$$\text{Field } \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$$

Equatorial Point:

$$\text{Field } \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{(-p)}{r^3}$$



General Point (θ from axis):

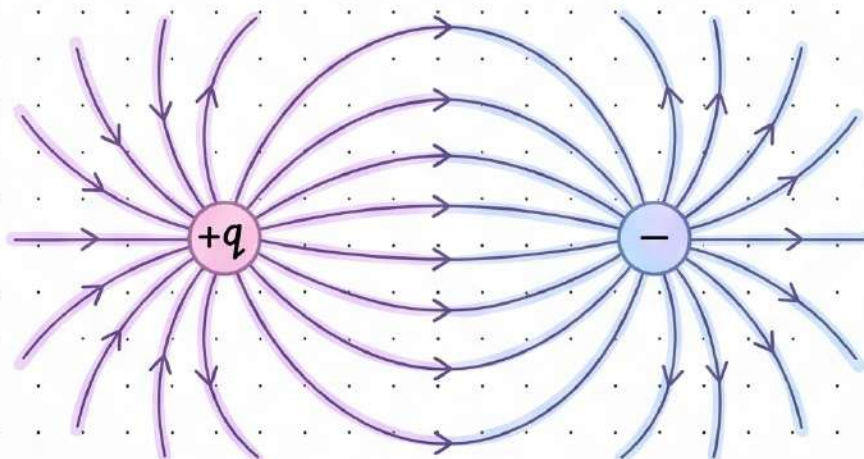
$$\text{Field } E = \frac{1}{4\pi\epsilon_0} \frac{p\sqrt{1+3\cos^2\theta}}{r^3}$$

Potential at general point:

$$V = \frac{1}{4\pi\epsilon_0} \frac{p\cos\theta}{r^2}$$

Electric Lines of Force Properties:

- They are imaginary, can never cross each other, and can never form closed loops.
- Lines of force end or start normally at the surface of a conductor.
- Tangent to the line of force at a point gives the direction of Electric Field.



Gauss's Law & Flux

Electric Flux and Gauss's Law:

Electric flux: $\phi = \int \vec{E} \cdot d\vec{s}$.

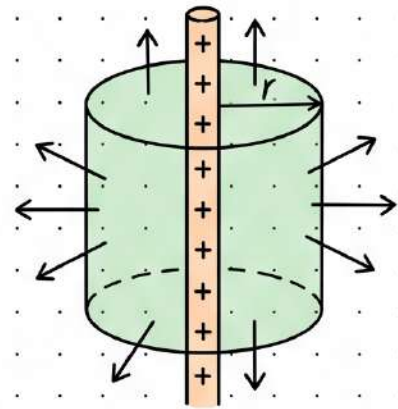
Gauss' Law (applicable only on closed surface): $\oint \vec{E} \cdot d\vec{s} = \frac{\sum q_{\text{enclosed}}}{\epsilon_0}$.

Applications (Electric Field $\vec{E}$):

Infinitely long line charge:

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{r}$$

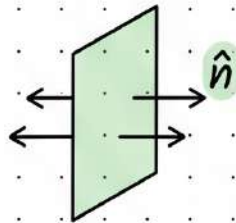
→ λ is linear charge density



Infinite non-conducting thin sheet:

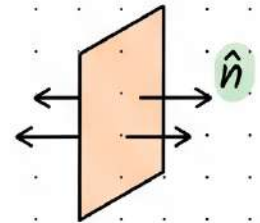
$$\vec{E} = \frac{\sigma}{2\epsilon_0} \hat{n}$$

σ is surface charge density



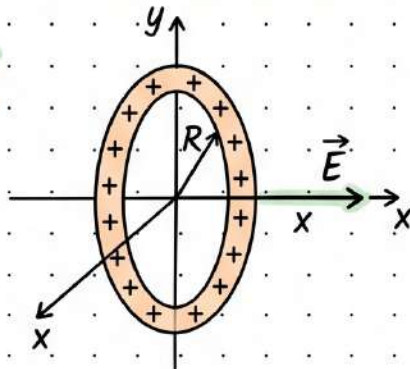
Infinite charged conducting thin sheet:

$$\vec{E} = \frac{\sigma}{\epsilon_0} \hat{n}$$



Uniformly charged ring (at axis distance x):

$$E = \frac{kQx}{(R^2 + x^2)^{3/2}}$$



Charged Long Conducting Cylinder (for $r \geq R$):

$$E = \frac{q}{2\pi\epsilon_0 r L}$$

Electrostatic Potential & Energy

Relation between E & V

$$\vec{E} = -\text{grad } V = -\nabla V$$

$$\Delta V = -\int \vec{E} \cdot d\vec{r}$$

For spherical and cylindrical symmetry:

$$E = \frac{-dV}{dr}$$

Electric Potential Energy

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

$$U = \frac{\epsilon_0}{2} E^2$$

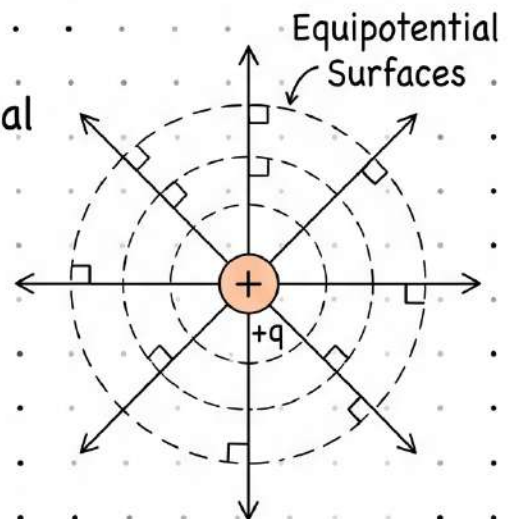
CETkit

Equipotential Surfaces

In an electric field the locus of points of equal potential is called an equipotential surface.

An equipotential surface and the electric field lines meet at right angles.

Material of conductors is an equipotential region.



Potential of a Uniformly Charged Ring

At center: $V = \frac{kQ}{R}$

At axis distance x : $V = \frac{kQ}{\sqrt{R^2 + x^2}}$

Potential & Field of Spheres

Uniformly Charged Hollow / Conducting Sphere

Outside ($r \geq R$)

Field $E = \frac{kQ}{r^2}$

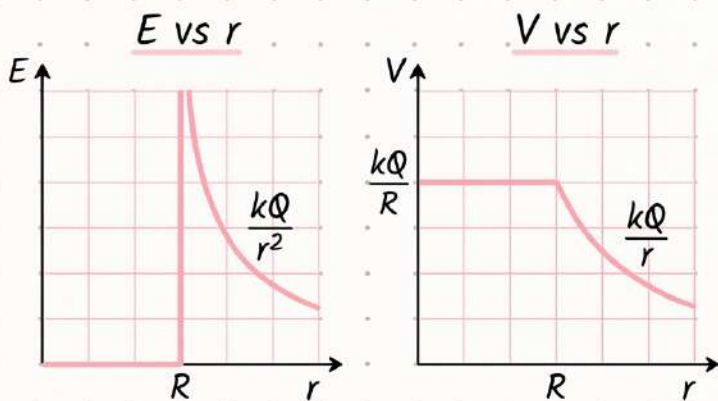
Potential $V = \frac{kQ}{r}$

Inside ($r < R$)

Field $E = 0$

Potential $V = \frac{kQ}{R}$

The sphere acts like a point charge placed at center for points outside the sphere.



Uniformly Charged Solid Non-Conducting Sphere

Outside ($r \geq R$)

Field $E = \frac{kQ}{r^2}$

Potential $V = \frac{kQ}{r}$

Inside ($r \leq R$)

Field $E = \frac{kQr}{R^3}$

Potential $V = \frac{kQ(3R^2 - r^2)}{2R^3}$

Inside the sphere, potential varies parabolically, and outside it varies hyperbolically.

Center Potential: $V_{\text{center}} = \frac{3}{2} V_{\text{surface}}$

